Infrastructure & Networking

#What is Infrastructure?

What is Infrastructure?

Infrastructure is the **complete set of hardware, software, network, and storage resources** that an organization needs to run its applications and services. Think of it like the foundation of a building — without it, nothing can run on top.

Compute Layer

Servers that process data. Contains CPU (brain), RAM (short-term memory). Every calculation, every web request, every database query — it all runs here.

Dell Technologies | HPE | Lenovo

Network Layer

Connects everything together. Switches, routers, NICs, and cables allow servers and devices to communicate with each other and the internet.

Cisco | Juniper | Arista

Storage Layer

Where data is kept permanently. Ranges from simple hard drives inside a server to massive enterprise SAN systems storing petabytes of data.

NetApp | Dell EMC | IBM

Virtualization Layer

Software that makes one physical server act like many. VMware ESXi, Microsoft Hyper-V, and KVM allow multiple virtual machines to share one physical box.

VMware | Microsoft | Red Hat

#Infrastructure Flow

[User Device: Laptop / Mobile]
↓
[NIC (Intel / Broadcom)] ← converts data to network packets
↓
[Network Switch (Cisco / Juniper)] ← routes traffic within datacenter
↓
[Router / Firewall (Cisco ASA / Palo Alto)]
↓
[Server (Dell PowerEdge / HPE ProLiant)] ← processes requests

↓
[Storage (NetApp / Dell EMC SAN)] ← fetches/saves data
↑
[Response sent back through the same path]

Real-World Example: When you open Swiggy on your phone, your request travels through your phone's NIC → your home router → the internet → Swiggy's datacenter switch → their web server → their database storage → and the menu data comes all the way back to your screen in under 500ms.

#NIC — Network Interface Card

📖 What is a NIC?

A NIC (Network Interface Card) is a hardware component — either a separate card that plugs into the motherboard or built directly into it — that allows a computer to connect to a network.

Every NIC has a **MAC address** (Media Access Control) — a unique 48-bit hardware ID burned into the card at the factory. Like your fingerprint — no two NICs in the world share the same MAC address.

🔬 How it works — Step by Step

1. **Application sends data**
   Your browser wants to load Google.com
2. **OS creates packets**
   Operating system wraps data into TCP/IP packets
3. **NIC converts signals**
   NIC converts digital packets → electrical signals (copper) or light pulses (fiber)
4. **Sends over cable**
   Signal travels through the cable to the network switch

📊 NIC Types — Comparison

TYPE	SPEED	USE CASE	INTERFACE	VENDOR EXAMPLE
1GbE NIC	1 Gbps	Home PCs, small offices	RJ45 (Ethernet)	Realtek, Intel I219
10GbE NIC	10 Gbps	Mid-range servers	RJ45 or SFP+	Intel X710, Broadcom 57810
25GbE NIC	25 Gbps	Modern data centers	SFP28	Mellanox ConnectX-5
100GbE NIC	100 Gbps	Hyperscale/cloud	QSFP28	Mellanox ConnectX-6, Intel E810
RDMA NIC (RoCE)	100+ Gbps	HPC, AI clusters	InfiniBand / QSFP	Mellanox (NVIDIA)

✅ Why use a dedicated NIC?

In servers, a dedicated high-performance NIC (like Intel X710) offloads network processing from the CPU — this is called **TOE (TCP Offload Engine)**. Less CPU work = more resources for your application.

📍 OSI Model — Where NIC Lives

- Layer 7 — Application (HTTP, FTP, DNS)
- Layer 6 — Presentation (SSL/TLS)
- Layer 5 — Session
- Layer 4 — Transport (TCP/UDP)
- Layer 3 — Network (IP, Router)
- **Layer 2 — Data Link ← NIC operates here (MAC address)**
- Layer 1 — Physical (cables, signals)

MAC Address Example: A MAC address looks like `00:1A:2B:3C:4D:5E`. The first 3 bytes (00:1A:2B) identify the manufacturer (Intel, Broadcom, etc.), the last 3 bytes are the unique device ID. You can find your NIC's MAC address with `ipconfig /all` on Windows or `ifconfig` on Linux.

#TCP vs UDP

🔵 TCP — Transmission Control Protocol

Connection-oriented. Before any data is sent, TCP does a *3-way handshake* to establish a reliable channel. Every packet is acknowledged (ACK). If a packet is lost, TCP re-sends it.

✅ Reliable ✅ Ordered delivery ✅ Error checking ❌ Slower (overhead)

Used for: Web browsing (HTTP/HTTPS), Email (SMTP/IMAP), File transfer (FTP), Online banking, SSH connections

🟡 UDP — User Datagram Protocol

Connectionless. UDP fires packets without any handshake. No acknowledgement, no retransmission. If a packet is lost — it's gone. But what you gain is *raw speed and low latency*.

✅ Very fast ✅ Low latency ❌ No reliability ❌ No ordering

Used for: Live streaming (IPL, YouTube Live), Online gaming (BGMI, CS2), VoIP calls (WhatsApp calls), DNS lookups, Video conferencing

🤝 TCP 3-Way Handshake — How Connection is Established

```
CLIENT SERVER Step 1: SYN ──────────────────────────────────────→ "I want to connect, Seq=100" Step 2:
SYN-ACK ←──────────────────────────────────── "OK, I acknowledge (ACK=101), Seq=300" Step 3: ACK
──────────────────────────────────────→ "Got it! ACK=301" == CONNECTION ESTABLISHED — Data transfer
begins == FIN ──────────────────────────────────────→ "I'm done" (4-way termination)
```

The 3-way handshake guarantees both sides are ready before data flows. This is why TCP is slower to start but reliable.

⚖️ Full Comparison — TCP vs UDP

FEATURE	TCP	UDP
Connection type	Connection-oriented	Connectionless
Speed	Slower (handshake + ACKs)	Faster (no overhead)
Reliability	✅ 100% guaranteed delivery	❌ Best effort, may drop
Ordering	✅ Packets arrive in order	❌ May arrive out of order
Error correction	✅ Retransmits lost packets	❌ No retransmission
Header size	20 bytes minimum	8 bytes only
Typical latency	~50–200ms (overhead)	~1–20ms
Best use	Banking, web, email	Gaming, live video, DNS

🏦 **Banking Example (TCP):** When you transfer ₹50,000 online, the bank cannot afford a single lost packet. TCP ensures every byte arrives correctly. One missed packet = corrupted transaction.

🏏 **IPL Streaming (UDP):** If one frame of the live cricket match drops, your eye won't even notice. UDP sends the next frame immediately — no waiting. Real-time > perfect quality.

#FCP — Fibre Channel Protocol

What is FCP?

📖 What is FCP?

Fibre Channel Protocol (FCP) is a high-speed **network protocol specifically designed for storage**. It runs over Fibre Channel networks (FC) and is used to connect servers to SAN (Storage Area Network) storage arrays in enterprise environments.

Unlike regular TCP/IP networks that handle all kinds of traffic, FCP runs on a **dedicated storage network** — separate from the regular data network. This means zero congestion, ultra-low latency, and guaranteed bandwidth for storage operations.

#FCP Architecture Diagram

[Server (Dell PowerEdge)]
↓
[HBA — Host Bus Adapter]
(like a NIC but for Fibre Channel — e.g., Emulex, Marvell QLogic)
↓

[Fiber Optic Cable]
(multi-mode or single-mode fiber — not copper!)
↓
[FC Switch (Brocade / Cisco MDS)]
(like an Ethernet switch but dedicated for FC traffic)
↓
[Storage Array (Dell EMC PowerStore / NetApp AFF)]
Block-level storage — the data lives here
═══ Separate dedicated FC fabric — isolated from regular network ═══

How FCP Works

1. **Server needs data**
   Database server needs a block of data from the storage array
2. **HBA sends FC frame**
   Host Bus Adapter creates a Fibre Channel frame (like an Ethernet frame but faster)
3. **FC Switch routes it**
   Brocade/Cisco MDS switch routes frame across the FC fabric to the right storage port
4. **Storage responds**
   Storage array sends the requested data block back at 16–32 Gbps

FCP vs iSCSI — When to choose

FEATURE	FCP	ISCSI
Speed	16–32 Gbps	10–25 Gbps
Latency	Sub-millisecond	1–5ms
Network	Dedicated FC fabric	Existing Ethernet
Cost	Very expensive	Affordable
Complexity	High	Low
Use case	Banking, hospitals	Mid-market

When to use FCP: Mission-critical workloads — HDFC Bank's core banking, hospitals' patient records, stock exchange trading systems where every microsecond matters.

Key Vendors

Brocade	Cisco MDS	QLogic/Marvell
FC SWITCHES (NOW BROADCOM)	FC SWITCH SERIES	HBA CARDS

#Server Components

CPU — Central Processing Unit

Server CPUs are different from desktop CPUs. They have **more cores (28–64+)**, support for more RAM, ECC (error-correcting) memory support, and are designed to run 24x7 for years.

Intel Xeon (Gold/Platinum) | AMD EPYC (Milan/Genoa)

Example: AMD EPYC 9654 — 96 cores, 3.4GHz, 384MB cache — used in hyperscale data centers

RAM — ECC Memory

Servers use **ECC RAM (Error-Correcting Code)** — special memory that can detect and fix single-bit errors automatically. A regular desktop losing 1 bit of RAM might cause a crash. A bank's server cannot afford that.

Samsung | SK Hynix | Micron

Why Servers ≠ PCs

FEATURE	PC	SERVER
CPUs	1	1–4 (multi-socket)
RAM type	DDR5 (no ECC)	DDR4/5 ECC RDIMM
Uptime goal	8 hrs/day	99.999% (5 nines)
Power supply	1x PSU	2x redundant PSU
Management	None	iDRAC/iLO (remote)
Form factor	Tower	Rack (1U/2U/4U)
NIC	1 GbE onboard	Dual 25/100 GbE

Server Form Factors

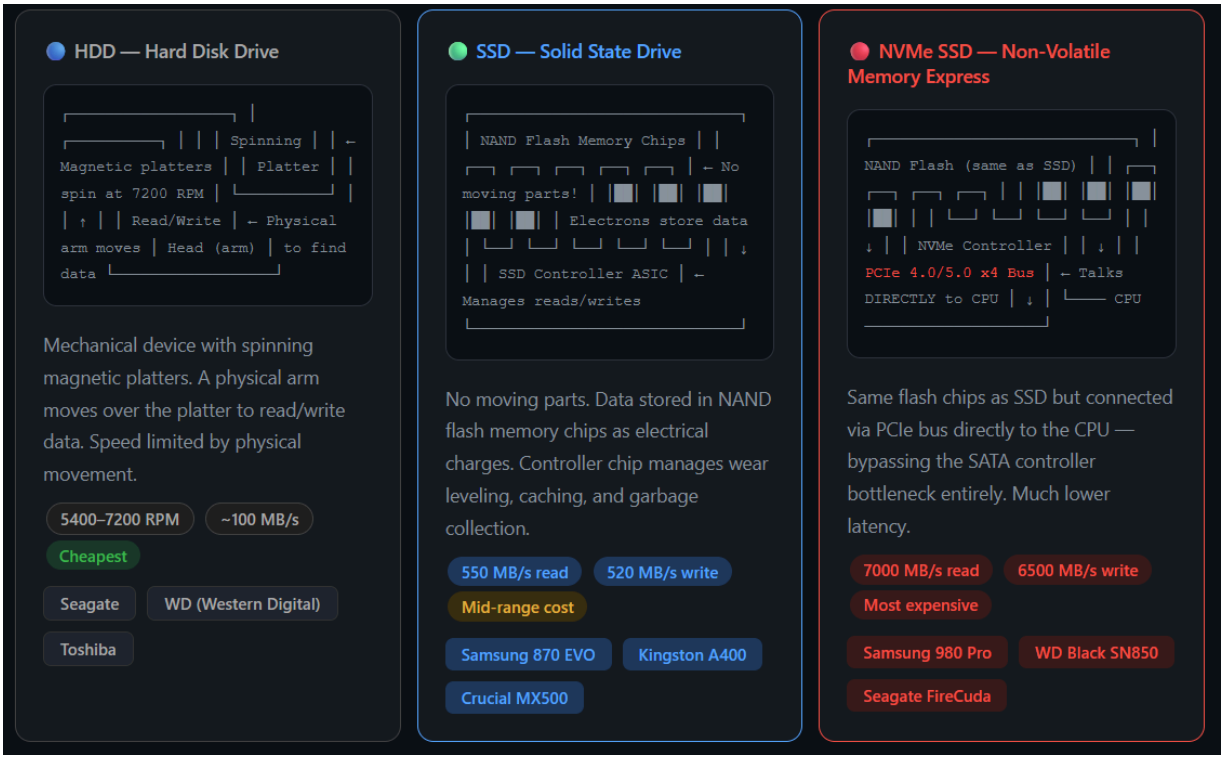
1U — **Rack server (1.75" height)**
e.g. Dell R650 — thin, fits 40+ in a rack

2U — **More drives/expansion**
e.g. Dell R750 — 3.5" height, more PCIe slots

BLD — **Blade server**
e.g. HPE BladeSystem — many blades in one chassis, shared power/cooling

#HDD / SSD / NVMe

Storage types — how they work inside, and when to choose each



Full Speed Comparison

Metric	HDD (7200 RPM)	SATA SSD	NVMe SSD (PCIe 4.0)
Sequential read	~150 MB/s	~550 MB/s	~7000 MB/s
Sequential write	~120 MB/s	~500 MB/s	~6500 MB/s
Random IOPS (4K)	~100–200	~90,000	~1,000,000
Latency	5–10ms	0.1ms	0.02ms
Cost per TB	~₹2,000	~₹6,000	~₹12,000
Lifespan	3–5 years	5–10 years	5–10 years
Best use	Backup, archival	OS, applications	Databases, video editing

Why is NVMe 13x faster than SATA SSD? SATA was designed in 2003 for spinning hard drives. It has a maximum queue depth of 1 command and 32 queue depth. NVMe was designed for flash — it supports 65,535 queues with 65,535 commands each. When a database fires thousands of simultaneous reads, NVMe handles all of them at once. SATA queues them up.

Interface Types

Physical connectors — storage and network interfaces explained

Storage Interfaces

INTERFACE	SPEED	CONNECTOR	USE CASE	WHO USES IT
SATA III	600 MB/s max	L-shaped 7-pin	Consumer SSDs, desktops	Home PCs, laptops
SAS (Serial Attached SCSI)	12 Gbps (1.5 GB/s)	SAS connector	Enterprise HDDs/SSDs	Dell/HPE servers
NVMe (PCIe 4.0 x4)	~7 GB/s	M.2 or U.2	High-perf workloads	Databases, AI
Fibre Channel	32 Gbps	LC fiber optic	SAN storage	Banks, data centers

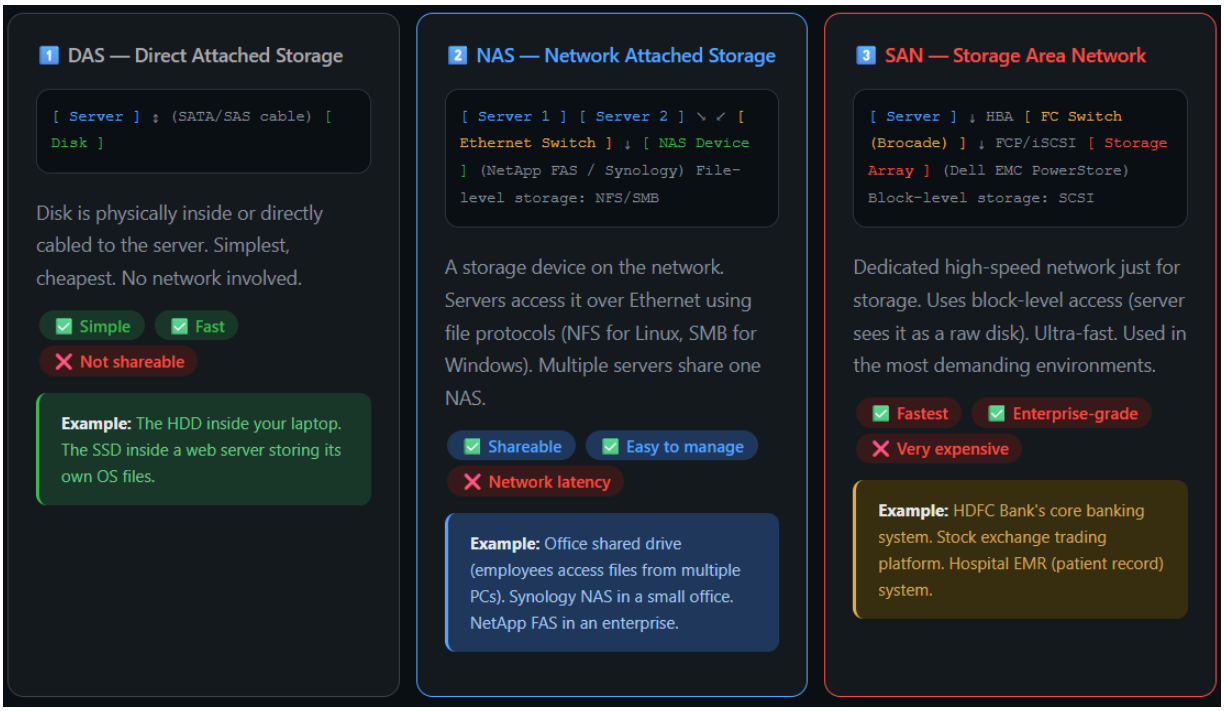
Network Interfaces

INTERFACE	SPEED	CONNECTOR	DISTANCE	USE CASE
RJ45 (Ethernet)	1–10 Gbps	8-pin RJ45	100m max	Office, home, servers
SFP+ (10G)	10 Gbps	SFP+ module	10km (fiber)	Data centers
QSFP28 (100G)	100 Gbps	QSFP28	40km (SMF)	Spine switches
InfiniBand (400G)	400 Gbps	QSFP-DD	Short range	HPC, AI clusters

SAS vs SATA — Why enterprises use SAS: SAS supports dual-port connectivity (two paths to the same drive — if one path fails, data still accessible). SATA is single-path only. In a server, a SAS drive can also be accessed simultaneously by two host adapters for load balancing.

SFP vs RJ45 at 10G: Both do 10 Gbps. RJ45 uses copper cable (cheap, but generates more heat, max 100m). SFP+ with fiber can run 10km. In data centers, fiber is preferred between switches because of distance and cleaner signal.

#Storage Architecture — DAS, NAS, SAN



DAS vs NAS vs SAN — When to Choose

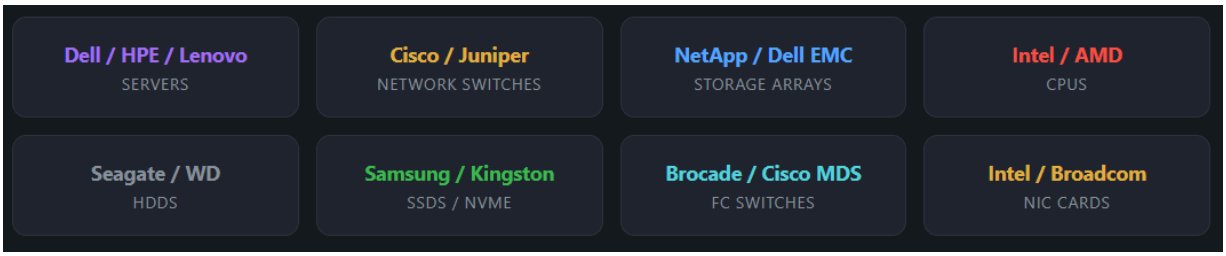
Criteria	DAS	NAS	SAN
Access type	Block	File (NFS/SMB)	Block (SCSI)
Shared access	❌ Single server only	✅ Multiple servers/users	✅ Multiple servers
Speed	Fast (direct)	Medium (ethernet)	Fastest (FC/iSCSI)
Cost	Lowest	Mid-range	Highest
Complexity	Simple	Moderate	Complex
Use case	Dev/test servers, OS	File shares, media	Databases, VMs
Vendor examples	Seagate, WD	Synology, NetApp	Dell EMC, IBM

#RAID Comparison

RAID Level	Min Disks	Speed	Can Survive	Usable %	Best For
RAID 0	2	2x	0 disk failures	100%	Speed-critical temp storage
RAID 1	2	1x	1 disk failure	50%	OS drives, critical data
RAID 5	3	~1.5x	1 disk failure	67–80%	File servers, general use
RAID 6	4	~1.5x	2 disk failures	50–75%	Large disks (>4TB) where

RAID Level	Min Disks	Speed	Can Survive	Usable %	Best For
					rebuilds take long
RAID 10	4	2x	1 per mirror pair	50%	Databases, VMs, high perf

#Why Each Technology Was Chosen

Scenario	Technology Used	Why This, Not Alternatives
Bank transaction	TCP	100% reliable delivery — can't afford to lose a payment packet
IPL live stream	UDP	1 dropped frame unnoticeable; speed matters more than perfection
DB storage	SAN (FCP)	Block-level, lowest latency, dedicated fabric — no congestion
Office file share	NAS (NFS/SMB)	Multiple users share files easily; cost-effective on Ethernet
Backup storage	HDD	Huge capacity per ₹; performance not critical for cold backup
Database server	NVMe SSD	1M IOPS needed for transaction processing — HDD/SATA can't handle it
Connectivity at scale	10/25GbE NIC	1GbE saturates with modern app traffic; 10G is the new minimum
Data protection	RAID 10	Database needs both speed and safety — RAID 10 gives both

#All Vendors